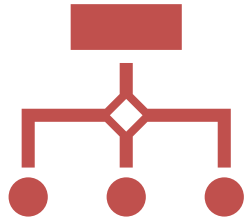


Networks and Systems Management



Monitoring and Performance
Filipe Sousa
FEUP – GRS/M.EIC

Monitoring

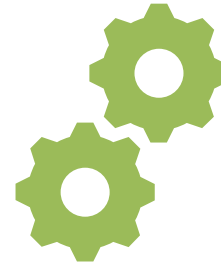


How is the network performing?

What quality is being offered?

Important for FCAPS - Fault, Accounting,
Configuration, Performance, Security

Follows application requirements and QoS?



Importance:

Ensures uptime and reliability

Detects bottlenecks and failures

Helps in capacity planning and security

Why to measure?



Short-term measurements

Identify faults, congestions, and attacks



Longer-term measurements

Traffic engineering, e.g. rerouting traffic or negotiating new agreements with peers
Upgrade link and device capacity



Accounting

To know how the network's clients are using the network and their usage

Types of measurements

Application and user-related measurements

- e.g. web page loading time

Device measurements

- CPU, memory, disk, link usage
- Temperature, fan speed, and other hardware measurements

Network measurements

- Traffic data – packet traces, flow data
- Latency
- Throughput
- Routing data

Key Network Performance Metrics

Latency: Delay in data transmission (measured in ms)

Bandwidth: Maximum data transfer rate (Mbps/Gbps)

Throughput: Actual data transfer rate

Packet Loss: Percentage of lost packets during transmission

Jitter: Variation in packet arrival times

Error Rate: Frequency of corrupted or dropped packets

Passive vs. Active network measurements

Passive - captures real network traffic

- Get a sense of the existing traffic in the network
- Have devices report how much traffic is going through (e.g. SNMP, NetFlow)
- Tap a link or copy traffic to the monitoring port (port mirroring)
- For measuring production traffic and its characteristics

Active - simulates traffic to measure performance

- Inject new, measurement packets in the network
- Get a sense of how the network reacts to these packets
- Including responses (e.g. ICMP req./reply for RTT)
- For measuring the properties of the network (delay, jitter, topology, etc.)

#1 Packet traces

- Motion-picture-like recording of everything that goes through the network
 - What, when, where, who (?), why (?)
- Raw data – powerful but hard to use
- Difficult to manage
 - Capture limitations (copy data at 1, 10, etc. Gbps scale)
 - Storage limitations (Gbps * minutes, hours, days = ?)
- Difficult to use and process
 - Not in a table-like format – would be easier
 - Can write processing rules to create tables – but only partial vision
 - AI and deep learning etc. to process traces (raw or features)

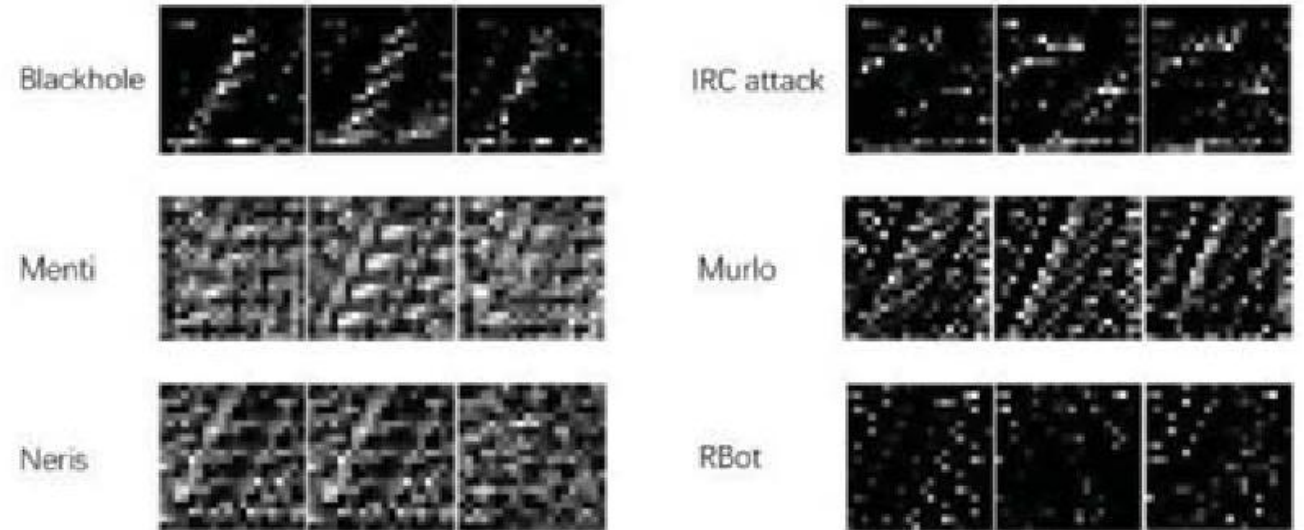
Packet trace usage example

Classify botnet traffic

First 400 IP payload bytes,
TCP flows

20x20 pixels, 0-255

Mostly good classification
results



Six samples of representation of the input data

Zhou, Z., Yao, L., Li, J., Hu, B., Wang, C., & Wang, Z. (2018). Classification of botnet families based on features self-learning under Network Traffic Censorship. 2018 Third International Conference on Security of Smart Cities, Industrial Control System and Communications (SSIC), 1–7.
<https://doi.org/10.1109/SSIC.2018.8556792>

#2 Traffic counters (SNMP, etc.)

Routers keep track of how much traffic goes through each link

- Packets, bytes
- Periodically every n minutes

Simple to use but limited in scope

- Graph of link usage
- Traffic matrix

Traffic matrices

Amount of data transmitted between every pair of nodes in the network

- rows and columns are the nodes in the network

Enterprise network, autonomous system:

- points of presence (PoP)
- links between PoPs – internal traffic
- links between PoPs and external AS devices – external traffic

Internal traffic matrix

- traffic between the PoPs in the AS

External traffic matrix

- traffic between PoPs and external AS's

#3 Flow measurements



IP flows

Source, destination IP address, and TCP/UDP ports (4 fields)

L3 header protocol (TCP, UDP, other)

Other info – ToS field, ??



Keeps record of traffic for each flow

Packet, byte count in each direction

Duration, first/last packet timestamps, TCP flags



Tradeoff

simpler to use than packet traces

more information than counters

simply opening a web page can generate dozens of TCP flows

Flow software

Netflow

- IETF IPFIX
- Export directly from routers
- Flow id, byte and packet counters, source/destination addresses and ports, duration, timestamps first and last packet
- Router looks up entries in flow cache, updates counters, or creates an entry
- Export flow records when idle (e.g. 15s)
- Expire flows with TCP RST/FIN, timeout, or flow cache full
- Sampling eases processing and memory, e.g. 1-N

tstat

- Open source for Linux, process trace file or directly from eth0
- 200+ flow features – IP, TCP, HTTP, TLS
- Domain name from DNS query, other fancy features
- Also UDP and other protocols – see tshark, scapy.

Active measurement tools



Depends on what you want to measure

ping

traceroute

owamp (rfc4656), twamp (rfc5357)

Iperf (rfc6349, iperf.fr)



Do it yourself

Raw sockets, scapy, TCP/UDP, application-level

What to do with measurement data?

Store for later query and processing

- Send to ELK, other big data storage
- Plot charts, do queries on past data, correlate between different data sources
- Build a historical dataset for learning AI models for different management tasks

Use immediately once data is generated

- Anomaly detection and diagnostic, security, traffic engineering, ...
- Apply static rules, use pre-trained AI model
- Online learning, update AI model

Performance and traffic engineering

Apply TE when congestion arises

- Admission control
- Policing and shaping (non-conforming traffic, limit rates)
- Queuing and scheduling policies

On the device

- Token bucket, fair queuing, etc.

On the network

- ATM (dead), IP (IntServ), label switching (DiffServ, MPLS/IP)

For protection

- Pre-routed alternative paths,
- Don't wait until link failure to start the routing algorithm
- Start sending through an alternative route when link failure detected

For load balancing

- Multipath TCP, DNS, reverse proxy

Common Network Issues and Troubleshooting

Issue	Possible Cause	Solution
High Latency	Poor routing, congestion	Optimize paths, QoS
Packet Loss	Network congestion, hardware failure	Check hardware, optimize traffic
Jitter	Unstable network paths	Improve network config
Slow Throughput	Bandwidth limitations, interference	Upgrade hardware, manage traffic

Key Takeaways



Network monitoring is essential for maintaining performance and security



Different monitoring tools serve various needs (e.g., Wireshark, Nagios)



Optimizing performance involves QoS, load balancing, and bandwidth management



Security monitoring is crucial to prevent attacks and downtime



Regular monitoring and proactive troubleshooting improve network efficiency

Further Reading & References

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- Tanenbaum, A. & Wetherall, D. (2020). Computer Networks. Pearson.
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